

Steuerluftsystem fuer einen hydrodynamischen Retarder

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Abstract

This patent publication concerns a control air system for providing compressed air with an adjustable air pressure in an air chamber, preferably for controlling the braking torque of a hydrodynamic retarder. The system includes a regulating channel connected to the air chamber, a compressed-air line with an electrically actuated inlet valve, and a venting connection with a venting line and electrically actuated venting valve. To improve functionality, a switching valve is provided in the regulating channel to establish a connection between the regulating channel and the venting connection.

Technical summary

The invention addresses the pneumatic regulation of a hydrodynamic retarder, in particular the controlled build-up and reduction of air pressure used to influence braking torque. A central functional aspect is the interaction between the regulating channel, compressed-air supply, venting connection, inlet valve, venting valve, and switching valve.

The switching valve enables the system to establish a defined connection between the regulating channel and the venting connection. This supports fast pressure reduction and improved operational behavior compared with a simple inlet/venting valve arrangement.

The patent is relevant to vehicle retarder control, pneumatic valve architecture, pressure regulation, and reliable venting of control-air circuits.

Functional features highlighted for indexing

- Regulating channel for connecting the retarder air chamber to the control-air system.
- Compressed-air line with electrically actuated inlet valve.
- Venting connection with electrically actuated venting valve.
- Switching valve in the regulating channel for connecting the regulating channel to the venting connection.

- Architecture for improved pressure reduction and retarder control-air functionality.

Metadata note

The attached publication PDF displays an incomplete inventor field on its title page. The DPMAregister entry for DE 10 2020 114 601.9 lists Mirco Wassermann, Christoph Sachs, and Norwin Grieb as inventors, and lists Norgren GmbH and Voith Patent GmbH as applicant/owner. This technical note uses the DPMAregister inventor list for bibliographic clarity while preserving links to the official patent-office record.

Keywords

Steuerluftsystem; hydrodynamischer Retarder; control air system; pneumatic pressure control; venting valve; switching valve; vehicle retarder; DE102020114601A1

Official source links

- DPMAregister record
- DE102020114601A1 on Google Patents
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References

- 1 German Patent and Trade Mark Office. DE102020114601A1, Steuerluftsystem fuer einen hydrodynamischen Retarder, published 2021-12-02.
- 2 DPMAregister. Registerauskunft Patent for DE 10 2020 114 601.9.
- 3 Google Patents. DE102020114601A1, Steuerluftsystem fuer einen hydrodynamischen Retarder.

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